

JACK BATES

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RESEARCH INTERESTS

What happens to accountability inside organisations when automated systems take over judgment. In the systems I've built, formal safeguards — approval gates, escalation rules, scoring thresholds — sat downstream of where the judgment was actually made, raising the question of whether they do the work we claim, or mainly make the process legible enough to sign off on. As automation expands, authority moves first; understanding erodes behind it, and the second is harder to reverse. I'm interested in whether the feedback loops needed to correct that — rather than merely document it — can still be built while the systems are simple enough to specify what correcting them means.

EDUCATION

University of Galway — BSc Business Information Systems 2023–2027

First Class Honours — Year 1 (71%), Year 2 (75%) · Academic Scholarship (Years 1 & 2)

- **Notable modules:** Advanced Database Technologies, Cybersecurity, Data Networks & Communications, Information Systems Technology, Advanced Application Development.

TBS Education, Barcelona — Bachelor of Management (Data Science for Business) 2025–2026

Double-degree exchange year.

- **Notable modules:** Programming, Big Data Modelling, Data Management & Visualization, The Business Aspect of AI, Econometrics, Data Governance.

Coláiste Choilm — Leaving Certificate 601 points

APPLIED RESEARCH & ENGINEERING EXPERIENCE

EUROCONTROL — Power Platform & AI Solutions Engineer Brussels · Apr–Aug 2026

Skyline Traineeship. Designed and deployed a suite of production AI agents for the Network Management Directorate.

- **Email triage & ticketing pipeline** — layered architecture over a shared service-desk mailbox: deterministic filters and request-type classification run first, an exception screen routes ambiguous mail to humans, and only clean cases reach the reasoning agent, which performs knowledge-base retrieval and returns a structured ticket / answer / route / discard decision. Evaluated over 25,000+ emails spanning ~15 months; measured ~40% of inbound volume as auto-ticketable and identified documentation coverage, not classification accuracy, as the binding constraint. Live in production.
- **Release Intelligence Agent** — ingests Microsoft 365 / Azure roadmap updates through a custom MCP server, assesses organisational impact against an internal knowledge base, and routes communications by priority band with human approval gating on high-impact items. Scoring is deliberately split from judgement: the model rates across seven weighted dimensions, deterministic code applies externally editable weights to compute the band, so any decision can be reconstructed and retuned without re-prompting. 800+ updates processed; ~8 hrs/week of manual monitoring displaced.
- **ITSD Help-Desk Agent** — retrieval assistant for service-desk staff, integrated with ServiceNow. Effort concentrated on constraining assertion rather than expanding retrieval: scope validation rejects near-miss articles with a stated reason, historical cases never substitute for documented procedure, and unanswerable queries escalate rather than produce a plausible guess.

Client Solution Architects — Shadowing Assistant. Digital solution implementation and client requirement workflows. 2020

PROJECTS

- **Writing-Voice LoRA Fine-Tune** — fine-tuned Llama-3.2-1B-Instruct (rank-32 LoRA, Unsloth) on 51 hand-curated examples of my own writing to test whether stylistic register transfers at small data scale, evaluated against the base model on held-out prompts. Style transferred; facts did not, and the fine-tune degraded below baseline out of distribution.
- **Healthcare Market Access Modelling (hackathon)** — classification, regression and clustering over 10,000+ observations across twelve European markets to model reimbursement approval and market share; applied multicollinearity diagnostics and class-imbalance handling. Found access conditions, not price, to be the dominant driver of adoption.
- **Personal Knowledge System** — agent-driven note system (Obsidian + Claude Code) turning raw notes and reading into a self-organising interlinked corpus — auto-tagging, cross-linking and flagging orphaned references.

SKILLS & CERTIFICATIONS

Technical: Python · SQL · R · Java · AI agent design (Copilot Studio, MCP connectors & tools) · LangGraph · Machine learning · MongoDB · Power BI

Certifications: ISO 8000 Master Data Quality Manager (ECCMA) · Enhance Agents with Autonomous Capabilities (Microsoft Applied Skills)

INTERESTS

Philosophy · Cybernetics · History

REFEREES

Available upon request.